Aswin D Menon

Master of Science | Aerospace Engineering
University of Michigan, Ann Arbor

aswin-d-menon.github.io

Research Interests

Safety-Critical Control Systems | Autonomous Systems | Robotics | Motion Planning & Navigation

Education

Degree	Institute	CGPA	Year of Graduation
Master of Science	University of Michigan, Ann Arbor	4.0/4.0 (current)	2027 (expected)
Bachelor of Technology	Indian Institute of Technology, Kharagpur	8.77/10	2025

Relevant Coursework

- **Control & Robotics:** Model Predictive Control | Online Learning for Control | Motion Planning and Control | Non-Linear Systems and Control | Math for Robotics | Embedded Control Systems | Robotics | Systems Laboratory
- **Dynamics & Guidance:** Aerospace Navigation and Guidance | Space Dynamics | Intermediate Dynamics | Flight Mechanics
- **Flight Controls:** Flight Vehicle Controls | Flight Stability & Control | Automatic Control of Aircraft | Helicopter Engineering

Publications

1. T. Kim, **A. D. Menon**, A. Trivedi, and D. Panagou, "Backup-Based Safety Filters: A Comparative Review of Backup CBF, Model Predictive Shielding, and gatekeeper," *arXiv preprint arXiv:2604.02401*, 2026.

2. T. Kim, R. H. Kumar, **A. D. Menon**, T.-H. Lin, and D. Panagou, "SEAMLiS: Visibility-Aware Safety for Perception-Limited Multi-Robot Exploration," *arXiv preprint arXiv:2607.09959*, 2026.

3. N. K. Singh, S. Pal, **A. D. Menon**, and S. Hota, "Impact Angle Constrained Guidance Laws for Stationary Target Interception in 3D Space," in *AIAA SCITECH 2026 Forum*, 2026, p. 2348.

Experience

- **Graduate Student Instructor (GSI) | AEROSP 470 | Prof. Ilya Kolmanovsky** *September 2026 - Present*
  - Assisting with course, exam, and homework logistics and coordinating grading for the **Control for Aerospace Vehicles** course
- **Graduate Research | Distributed Autonomous Systems and Control Lab | Prof. Dimitra Panagou** *September 2025 - Present*
  - Formulated a unified comparative framework to evaluate **Backup CBF**, **Model Predictive Shielding**, and **gatekeeper** filters
  - Implemented **SEAMLiS**, a modular execution-layer safety filter for decentralized **sensing-limited multi-robot exploration**
  - Developed an **Event-Triggered gatekeeper** to significantly reduce **computational costs** while preserving recursive feasibility
- **Engineering Intern | ADAS Features Algorithm Development Team | Magna Electronics Inc.** *May 2026 - August 2026*
  - Optimized the **velocity controller** of the **Trained Parking Assist (TPA)** feature utilizing in-vehicle testing and log analysis
  - Improved the **curvature-based slowdown** of the **ACC velocity controller** and validated the behavior in **CarMaker** simulations
  - Integrated the **TPA** model with other low-speed features and validated back-to-back feature behavior in vehicle & simulation
- **Undergraduate Research | Unmanned Aerial Vehicles Lab | Prof. Sikha Hota** *August 2024 - July 2025*
  - Implemented **Biased PNG**, **Two-Stage**, and **Time-to-go PNG** for 2D intercepts considering **acceleration and FOV constraints**
  - Adapted **multi-planar** guidance laws for **3D interception**, successfully eliminating lateral acceleration profile discontinuities
  - Designed a **2-planar 3-Staged 3D** Impact Angle Control Guidance (**IACG**) law that can achieve any terminal impact direction
- **Research Intern | Control Law Group | RWR&DC | Hindustan Aeronautics Ltd** *May 2024 - July 2024*
  - Analyzed helicopter **AFCS** and integrated a **Pitt-Peters** dynamic inflow model to improve the transient response predictions
  - Validated the simulated transient responses against flight test data for **collective dump** inputs to identify future developments

Projects

- **Academic Project | Model Predictive Control** | **Prof. Ilya Kolmanovsky** *Jan 2026 - April 2026*
  - Engineered a **constrained MPC** using a **dual projected gradient QP** solver to achieve offset-free yaw rate tracking for vehicles
  - Integrated soft constraints via **slack variables** to ensure robust control performance against the plant-model mismatches
- **Academic Project | Non Linear Control of an Inverted Pendulum** | **Prof. Inigo Incer** *Jan 2026 - April 2026*
  - Engineered a full-state feedback controller for a **nonlinear underactuated system** using dimensionless state transformations
  - Implemented a **Luenberger observer**-based dynamic compensator to estimate velocities and to verify closed-loop stability
- **Unitree Go2 High-Speed Trajectory Controller** | **Dimensional OS (Open Source)** *Feb 2026*
  - Upgraded the **control architecture** from a 10Hz reactive model to a 100Hz setup with a **ROS** adapter for high-speed tracking
  - Designed an **adaptive Pure Pursuit** and **PID Cross-Track controller** alongside a novel **velocity profiler** to stop lateral drifts
- **Academic Project | Online Learning for Control** | **Prof. Vasileios Tzoumas** *Sep 2025 - Dec 2025*
  - Implemented an **adaptive SSI-MPC** in **ROS2** utilizing Recursive Least Squares and predictors for unknown UAV dynamics
  - Integrated **Acados-based** solvers in a **Gazebo SITL** for high-speed multi-drone **trajectory tracking** against wind disturbances
- **Academic Project | Aerospace Navigation and Guidance** | **Prof. Giusy Falcone** *Sep 2025 - Dec 2025*
  - Engineered a discrete-time **Kalman Filte**r for accurate **state estimation** under a **Gaussian** process and measurement noise
  - Designed a computationally efficient **pursuit guidance law** for dynamic target interception in a high-speed engagement case
- **Aerial 2D Mapping and Estimation of Forest Area Cover** | **Prof. Sandeep Saha** *Jan 2024 - Apr 2024*
  - Automated **semantic segmentation** and large-scale **environmental perception** via a **U-Net CNN** utilizing aerial drone imagery
  - Constructed maps from **telemetry** in **Pix4Dmapper**; applied **EDSR** to generate **Digital Surface Models** to estimate elevation
- **Aerial Carbon Emission Monitoring System** | **Prof. Srinibas Karmakar** *Aug 2023 - Nov 2023*
  - Designed a drone-deployed embedded payload with **MQ7** and **MQ135** sensors, utilizing **nRF24L01** for real-time **RF** telemetry
  - Developed an **ascent time trajectory algorithm** and altitude **state estimation** via **BMP280** and dynamic atmospheric models

Technical Skills

- **Languages**: Python | MATLAB | C | Arduino | Visual Basic | LaTeX
- **Software**: ROS | Gazebo | RViz | IPG CarMaker | dSPACE ControlDesk | MATLAB/Simulink | Fusion 360 | Pix4Dmapper

Achievements

- **AIR 5855** (top **0.55%**) in **JEE Advanced** | **AIR 8532** (top **0.8%**) in **JEE Mains** | **Rank 28** (out of **70000**) in **KEAM** *(2021)*
- Awarded departmental transfer to Aerospace Engineering based on a strong freshman-year academic performance *(2022)*

Extra-Curriculars

- Won overall **gold** at the Inter IIT Cultural Meet, 2023; Secured **3rd** (2023) & **5th** (2024) for Inter-Hall Short Film Making
- Member of RadhaKrishnan Hall soccer team for Inter-Hall (2022 to 2025); **Finalists** (2025); **Quarterfinalists** (2023 & 2024)